Md. Abu Bokkor Shiddik

Researcher | Bachelor of Science in Statistics

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Link: Portfolio | LinkedIn | Google Scholar | ORCID | GitHub | Web of Science | Scopus ID

RESEARCH INTEREST

I have a strong interest in using data science, machine learning and explainable AI to improve global health, especially in disease modeling and climate–health decision making.

SKILLS

Machine Learning, Deep Learning & AI

Regression, classification, decision trees, random forest, gradient boosting (XGBoost, LightGBM, CatBoost), SVM, KNN, Naive Bayes, clustering, PCA, neural networks (ANN, CNN, RNN, LSTM, GRU)

Explainable AI & Model Interpretation

SHAP, LIME, Integrated Gradients, Layer-wise Relevance Propagation, feature importance analysis

Statistical, Spatio-Temporal & Health Analytics

Hypothesis testing, ANOVA, GLM, ARIMA/SARIMA, Bayesian inference, spatio-temporal modeling, hotspot analysis (Gi*), GIS (sf, raster), epidemiological & disease prediction modeling, climate–health analysis

Programming & Data Tools

R, Python, C, Java, SPSS, LaTeX, Git/GitHub, HTML

Research Practice

Scientific writing, systematic review, reproducible research

EDUCATION

March, 2022 -June, 2026	**Bachelor of Science in Statistics** Begum Rokeya University, Rangpur, Bangladesh **Session**: 2020-21; **Result**: CGPA: 3.72 out of 4.00
January, 2018 - December, 2019	**Higher Secondary Certificate (Science)-2019** Police Lines School and College, Rangpur, Bangladesh **Session**: 2017-18; **Result**: GPA: 5.00 out of 5.00
January, 2015 - December, 2016	**Dakhil (SSC) Certificate (Science)-2017** Pakuria Sharif Se. Fagil Madrasah, Rangpur, Bangladesh **Result**: GPA: 5.00 out of 5.00

CONFERENCE & WORKSHOP

International Conference on Applied Statistics and Data Science-2025, University of Dhaka

Abstract ID: P-145 & P-105

Presented research work: Spatio-temporal modeling of global tuberculosis prediction based on unified-machine learning framework

Workshop on 'Promotion of Research in Higher Education (10 march, 2025)

Institutional Quality Assurance Cell (IQAC), Begum Rokeya University, Rangpur.

CERTIFICATIONS

March, 2026	**Certificate of Publication** The certificate of publication for the article titled:District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning; *Trop. Med. Infect. Dis.* 2026, Volume 11, Issue 3, 73
March, 2026	**Certificate of Excellence; Research design** by Researcher Academy, ELSEVIER
23 March 2026	**Certificate of Appreciation in recognition of peer reviewer contributions for BMJ Global Health** by BMJ

16 February 2026	**Certificate of Appreciation in recognition of peer reviewer contributions for BMJ Public Health** by BMJ
January, 2026	**Certificate of Excellence; Technical writing skills** by Researcher Academy, ELSEVIER
December, 2025	**Certificate of Appreciation for presenting a research paper at the Poster session of the International Conference on Applied Statistics and Data Science (ICASDS)** by Institute of Statistical Research and Training (ISRT), University of Dhaka

COURSES AND TRAINING

January - March, 2026	**Scientific & Manuscript Writing Series** (**6 modules**: Gen AI use in the research workflow, interdisciplinary writing, case reports, identifying research gaps, LaTeX manuscripts, methods articles, review articles) by Researcher Academy, Elsevier
15 March, 2025	**Practical Time Series Analysis** An online course by The State University of New York
10 September, 2024	**Introduction to Climate Risk Informed Decision Analysis (CRIDA)** a course of study offered by UNESCO
January – Jun 2024	**Data Science & Machine Learning Specialization** (**6 courses**: Machine Learning with Python, Data Visualization with R, Regression for Car Mileage/Diamond Price, R for Data Science, Data Science Methodology, What is Data Science) by IBM
10 March, 2024	**Successful Presentation** An online course by University of Colorado Boulder
December 2023	**Research Methodology Series** (**4 courses**: Interdisciplinary Research Methodology, Research Methodologies, Understanding Research Methods, Introduction to Computer Programming) by RSBRU / Queen Mary University of London / University of London
24 December, 2023	**Introduction to Statistical Analysis: Hypothesis Testing** By SAS
November, 2023	**UNICEF Training Series** (**2 courses**: Education in Emergencies, Leadership for Planning and Decision-Making with Adolescents)
July, 2016-December, 2016	**Deploma in Computer Science** A six month training course by Bangladesh Computer Education (BCE)

RESEARCH PROJECT

Project Title: Exploring community knowledge, health belief and healthcare-seeking behaviours regarding Non-Communicable Diseases (NCD's) in Bangladesh

Funding Source: University Grant Commission; **Role**: Research Assistant (Start date: Oct.2025 & end date: Feb.2026) , Department of Statistics, Begum Rokeya University, Rangpur

Project Title: District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning

Description: A project submitted in partial fulfillment of the requirements for the degree of B.Sc. In Statistics, Begum Rokeya University, Rangpur.

RESEARCH & PUBLICATION

1. Rahman, M.S., **Shiddik, A.B.** Utilizing artificial intelligence to predict and analyze socioeconomic, environmental, and healthcare factors driving tuberculosis globally. *Sci Rep* 15, 13619 (2025). https://doi.org/10.1038/s41598-025-96973-w

2. Rahman, M. S., Amrin, M., & **Bokkor Shiddik, M. A.** (2025). Dengue Early Warning System and Outbreak Prediction Tool in Bangladesh Using Interpretable Tree-Based Machine Learning Model. *Health Science Reports*, *8*(5), e70726. https://doi.org/10.1002/hsr2.70726

3. Rahman, M. S., & **Shiddik, M. A. B**. (2025). Explainable artificial intelligence for predicting dengue outbreaks in Bangladesh

using eco-climatic triggers. *Global Epidemiology*, 100210.

4. Rahman, M. S., & **Shiddik, M. A. B.** (2025). Unraveling global malaria incidence and mortality using machine learning and artificial intelligence–driven spatial analysis. *Scientific Reports*, *15*(1), 28334.

5. Rahman, S., & **Shiddik, A. B**. (2025). Reflections on explainable artificial intelligence for predicting dengue outbreaks in Bangladesh. *Global Epidemiology*, 100230.

6. Rahman, M. S., & Shiddik, M. A. B. (2025). Data-Driven Dengue Prevention Strategies in Bangladesh using Explainable Artificial Intelligence and Causal Inference. *International Journal of Statistical Sciences*, *25*(2), 101-113.

7. Rahman, M. S., & **Shiddik, M. A. B.** (2025). Leveraging explainable artificial intelligence and spatial analysis for communicable diseases in Asia (2000–2022) based on health, climate, and socioeconomic factors. *International Journal of Health Geographics*, *24*(1), 45.

8. Rahman, S. and **Shiddik, A.B.**, 2026. Bayesian spatiotemporal modelling of neonatal, infant and under-5 mortality (2000–2022) in 41 Asian countries: a population-level observational study. *BMJ open*, *16*(2), p.e108632.

9. Shiddik, M.A.B., Toshi, F.Z., Yesmin, S. and Rahman, M.S., 2026. District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning. *Tropical Medicine and Infectious Disease*, 11(3), p.73.

10. Rahman, M. S., **Shiddik, M. A. B.**, & Chowdhury, A. H. (2026). Spatiotemporal patterns of climate-sensitive vector-borne diseases in Bangladesh: leveraging machine learning and spatial regression for intervention strategies. BMC medicine.

11. Shiddik, M.A.B., 2026. Global Prediction of Dengue Incidence Using an Explainable Artificial Intelligence-Driven ConvLSTM Integrating Environmental, Health, and Socio-Economic Determinants. Health Science Reports, 9(4), p.e72280.

12. Shiddik, M.A.B., 2026. Explainable Artificial Intelligence in Healthcare: Current Landscape, Challenges, and Future Directions. Health Science Reports, 9(3), p.e72172.

13. Rahman, M. S., & **Shiddik, M. A. B.** (2026). Stillbirth, Neonatal, and Child Mortality in Bangladesh: Progress and Persistent Public Health Challenges. Public Health Challenges, 5(3), e70312.

REVIEWER SERVICE

- **Global Public Health** (2 Article in 2026)
- **BMJ Global Health** (2 Article in 2026)
- **BMJ public health** (2 Article in 2026)
- **Health science reports** (1 Article in 2026, 2 Article in 2025)
- **Journal of mathematics and statistics** (1 Article in 2025)
- **PLOS Digital Health** (8 Article in 2026, 6 Article in 2025)

DIGITAL PROJECTS AND TOOLS

whatdis() — R distribution detector
Developed whatdis(), a custom R package for checking and visualizing data distribution. Code and documentation available on GitHub: https://github.com/abubokkorshiddik/R-distribution-detector

Dengue analytics dashboard, Bangladesh
Built an interactive Dengue dashboard that shows patterns and insights for district level dengue cases and early warning modeling in Bangladesh. Live demo:
https://abubokkorshiddik.github.io/dengue-dashboard

Systematic review dashboard
Built an interactive systematic review dashboard for searching, screening, trend visualization, filtering, and result export. Live demo: https://abubokkorshiddik.github.io/research-explorer

ACADEMIC CONTENT AND OUTREACH

Created educational video tutorials on Basic Statistics, R and Python for data analysis
YouTube channel: https://www.youtube.com/@abubokkorshiddik.abs798

Shared public health and data science resources to promote research skills among students and early-career researchers

REFERENCE

Dr. Md. Siddikur Rahman Associate Professor Department of Statistics, Begum Rokeya University,Rangpur, Bangladesh. E-mail: siddikur@brur.ac.bd Tel: +8801711339573	**Dr. Md. Bipul Hossen** Associate Professor Department of Statistics, Begum Rokeya University,Rangpur, Bangladesh. E-mail: mbipu@brur.ac.bd Tel: +8801723-986994